

Universum Inspired Graph Contrast Learning for Autism Spectrum Disorder Identification

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Abstract—The functional brain network (FBN) has recently emerged as a pivotal biomarker in the diagnosis of autism spectrum disorder (ASD). Meanwhile, graph convolutional networks (GCNs) represent a promising strategy for the concurrent extraction of FBN features and the facilitation of ASD identification. Unfortunately, the efficacy of existing GCN methods is often limited by their reliance on extensive annotated datasets. To address this limitation, we propose the Universum Inspired Graph Contrast Learning (Uni-GCL) framework, a novel approach to brain disease analysis that capitalizes on the generation of Universum samples for the pre-training of the GCN model. Specifically, we use a Mixup technique to generate universum samples and design a universum-inspired contrastive loss function. This methodology necessitates only a modest number of samples for finetuning the pre-trained model, effectively reducing data requirements while simultaneously boosting model performance. To systematically assess the performance of Uni-GCL, we conducted rigorous experiments on the Autism Brain Imaging Data Exchange (ABIDE) initiative for ASD identification. The results demonstrate that Uni-GCL, augmented with universum samples, yields compelling outcomes, underscoring its potential as a robust method in the diagnostic arsenal for ASD.

Index Terms—Functional Brain Networks, Autism Spectrum Disorder, Graph Convolutional Networks, Universum Learning, Graph Contrast Learning

I. INTRODUCTION

Autism Spectrum Disorder (ASD) is a psychological disorder that typically emerges during early childhood [1][2][3]. Currently, there is no definitive cure for ASD[4]. However, early intervention, coupled with educational support and therapeutic care, can markedly enhance the functionality and overall quality of life for affected individuals [5]. Consequently, accurate and timely diagnosis of ASD is of paramount

importance.

Recently, there have been some works combining brain networks with machine learning techniques for disease diagnosis. For instance, Pan et al. used graphical neural networks for neuroimaging diagnosis[6], Xu et al. studied large-scale brain networks in ASD patients[7], Li et al. proposed a multi-level dynamic brain network joint learning architecture based on GCN[8], and Alamdari et al. investigated the interactions between brain functional networks[9]. However, the efficacy of these studies is often constrained by their reliance on extensively annotated datasets, and high-quality annotated data is difficult to obtain and costly. This has prompted the exploration of strategies to diminish the reliance on large sample sizes.

In response, this study presents a Universum-Inspired Graph Contrastive Learning (Uni-GCL) framework, a novel approach that pre-trains a Graph Convolutional Network (GCN) using Universum samples. The key idea is to create Universum samples using Mixup techniques and to develop a new universum loss function for self-supervised learning, thereby reducing the need for annotated training data. The paper also employs a dual-layer fMRI enhancement strategy that increases sample size by enhancing blood-oxygen-level-dependent (BOLD) signals. Additionally, it constructs three parallel spatiotemporal graph convolutional networks (ST-GCNs) for fMRI feature extraction and performs unsupervised contrastive learning. Experiments on the ABIDE dataset show that Uni-GCL is more effective in identifying brain diseases such as autism than previous methods. The main contributions of this article include:

- We have developed an unsupervised contrastive graph learning framework that effectively addresses the need for GNN to analyze fMRI and diagnose brain diseases in the context of scarce medical data.
- We introduced a rational two-stage fMRI enhancement strategy that preserves the functional connectivity between ROIs by augmenting the original BOLD signal in resting-state fMRI.

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- We are using Universum samples for pre-training GCN for the first time to improve model generalization and reduce reliance on large datasets.

Inspired by the universum learning method, our model introduces universum data as strong negative samples to assist in classification tasks. Different from traditional self-

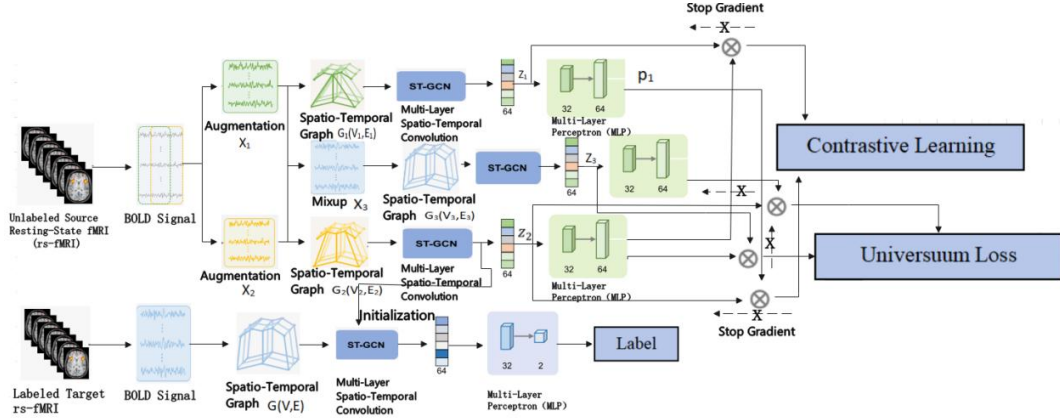


Fig 1. The framework of our Uni-GCL

II. RELATED WORK

In this section, we will briefly introduce the Mixup strategy, as well as universum contrastive learning.

A. Mixup strategy

Mixup is a data augmentation technique that generates new training samples through linear interpolation to improve the generalization ability of deep learning models[10]. Specifically, mixing entails blending two data samples with their respective labels using a mixing ratio[11]. This ratio is a random variable uniformly distributed within the interval [0,1]. For instance, the computation of the mixed sample and label is as follows:

$$x = \lambda x_1 + (1 - \lambda)x_2 \quad (1)$$

$$y = \lambda y_1 + (1 - \lambda)y_2 \quad (2)$$

Mixup combined with cross entropy loss simplifies model tuning, enhances the handling of outliers and noise, and improves model generalization and robustness[12]. In practice, Mixup can be combined with other enhancement techniques or regularization methods to enhance performance. This study innovatively uses Mixup to generate universum samples as negative samples for pre-training the model, increasing data diversity and improving generalization ability[13].

Unlike traditional Mixup strategies that directly train models, this paper uses the Mixup strategy to generate universum samples, treating them as negative samples. The purpose of doing so is to separate the universum data from all known categories and pre-train the model, which can enhance the diversity of the data and the generalization ability of the model.

B. Universum Learning

Universum learning is an emerging method in the field of machine learning proposed by researchers such as Vapnik, which improves model generalization ability by considering all possible patterns[14][15]. It has the characteristics of global assumptions and regularization, and aims to learn all potential data patterns rather than just visible data[16].

supervised learning, universum samples provide an additional source of information that can help the model identify which features are key to distinguishing between classes in the training data and which features may be noise or irrelevant, enhancing the model's robustness to distribution shifts.

C. Discussion

Existing machine learning methods often require a large amount of annotated data, which is particularly difficult in the field of medical imaging because obtaining and annotating data is both expensive and time-consuming. In order to alleviate the reliance on annotated data, this study adopts the Mixup strategy to perform linear interpolation between samples for generating universum samples. This sample forces the model to learn more discriminative feature representations and better generalize to unseen data. By considering all possible data distributions instead of just limited to the currently visible data, universum samples significantly improve the accuracy of the model for ASD diagnosis and reduce the need for a large amount of annotated data. This method is consistent with the universum learning approach proposed by Vapnik et al., which aims to improve the model's generalization ability by utilizing all possible data patterns.

III. DATA PREPROCESSING

A. Data preprocessing

In this paper, we have selected the ABIDE dataset for our experiments. For ABIDE, we have chosen the NYU site, the UM site and the LEUVEN site to validate our model. These three sites include 79 autism spectrum disorder (ASD) patients and 105 healthy controls (HC), 68 ASD patients and 77 HC patients, and 29 ASD patients and 35 HC patients, respectively. At the NYU site, we collected resting-state functional magnetic resonance imaging (rs-fMRI) data using an Allegra scanner with a magnetic induction strength of 3 Tesla. Each MRI scan had a repetition time (TR) of 2000 ms, an echo time (TE) of 15 ms, a voxel size of $3.0 \times 3.0 \times 4.00 \text{ mm}^3$, number of slice is 33, a thickness of slice is 4.0 mm, and a flip angle of 90° . At the UM site, rs-fMRI data were collected using a GE Signa

scanner with a magnetic induction strength of 3 Tesla. Each MRI scan had a TR of 2000 ms, a TE of 30 ms, a voxel size of $3.44 \times 3.44 \times 4.00 \text{ mm}^3$, number of slice is 40, a slice thickness of 3.0 mm, and a flip angle of 90° . At the LEUVEN site, data were collected using a Philips MR scanner with a magnetic induction strength of 3 Tesla. Each MRI scan had a TR of 1667 ms, a TE of 33 ms, a voxel size of $3.59 \times 3.59 \times 4.00 \text{ mm}^3$, number of slice is 32, a thickness of 4.00 mm, and a flip angle of 90° . All fMRI data included in the compliance database were preprocessed using DPARSF.

In the initial stage, resting-state functional magnetic resonance imaging (fMRI) data undergoes several standard preprocessing steps. These steps involve correcting slice timing, adjusting for head motion, performing regression analysis to remove interference signals, normalizing images to a standard brain space, smoothing images to enhance signal-to-noise ratio, and applying a bandpass filter to isolate signals within a specific frequency range. Following this, the brain is divided into 116 regions of interest (ROIs) using the AAL atlas, and the average time series for each ROI is extracted.

IV. UNIVERSUM INSPIRED GRAPH CONTRAST LEARNING

To reduce the reliance for data and improve model performance, we propose a Universum-Inspired Graph Contrastive Learning method, which mainly includes the following parts, as detailed below:

A. Bi-level fMRI augmentation

To improve the generalization capability of deep learning models in fMRI data analysis and reduce overfitting, this study introduces a data augmentation technique utilizing the original BOLD signal. This approach employs a window-slicing strategy that trims the initial and final segments of the original time series to generate new samples, thereby extending the original BOLD time series. This method preserves functional connectivity information, preventing direct enhancement of 4D functional magnetic resonance scanning[17]. Importantly, this method effectively prolongs time series data without introducing extra noise or altering time dependencies.

Specifically, $X = (x_1, \dots, x_N)^T \in R^{N \times T}$ as a subject's rs-fMRI time series, where $x_n = (x_{1,n}, \dots, x_{T,n}) \in R^T$ denotes the mean BOLD signal intensity for the n-th region of interest (ROI), T corresponds to the count of time points in the unaltered BOLD signal, while N=116 represents the total number of ROI. By selecting a slice length S, the initial and final S time points of the time series can be truncated to generate two new BOLD signal samples. Mathematically, these enhanced BOLD signals can be represented as $X_i = \tau_i(X) \in R^{N \times S} (i = 1, 2)$, which are subsequences of the original time series extracted from the front S column and the back S column using a window-slicing strategy, respectively.

B. Parallel fMRI feature extraction module

To enhance the generalization ability of the model in fMRI data analysis, this study employed an unsupervised contrastive learning framework. This framework comprises three parallel networks, all utilized for feature extraction. Initially, the enhanced BOLD signal and Universe signal are combined to

form a spatiotemporal graph, which is then input into three spatiotemporal graph convolutional networks (ST-GCN) to extract the spatiotemporal features of the data[18]. Subsequently, a multi-layer perceptron (MLP) with two fully-connected layers of 32 and 64 neurons each is used to process the features from each branch. The stop gradient operation is applied in every ST-GCN to prevent model interference. By comparing loss functions, the similarity between different representations of the same individual is enhanced, enabling unsupervised learning of fMRI information representation without class labels.

To simulate the functional connectivity network of the brain, a spatiotemporal map was constructed for each view. $G_i(V_i, E_i)$ as a spatiotemporal graph was constructed for an enhanced view $X_i (i = 1, 2)$. A set of nodes/regions of interest (ROIs) defined at S time points as follows:

$$V_i = \{v_{t_i,n} | t_1 = 1, \dots, S; t_2 = T - S + 1, \dots, T; n = 1, \dots, N\} \quad (3)$$

E_i is defined as a collection of edges that signify the interconnections among auxiliary nodes across both temporal and spatial domains. In more detailed terms, for every region of interest (ROI) and each moment in time, an edge in the temporal graph links the pertinent node to the same ROI at successive time points. Concurrently, at a given time point and ROI, the spatial graph's edges join all nodes present at that same moment, with the weights of these edges being determined by the Pearson correlation coefficients between the different ROIs.

In the ST-GCN model, for the purpose of deriving brain connectivity patterns and temporal information from fMRI time series data, this study employs spatiotemporal graph convolution (STGC) on every input spatiotemporal graph. To define convolution on G_i , we use $f_{in}(v_{t_i,n})$ to represent the node features $v_{t_i,n}$ and $B(v_{t_i,n})$ to represent the spatiotemporal domain of points $v_{t_i,n}$ as follows:

$$B(v_{t_i,n}) = \{v_{q,m} | d(v_{t_i,m}, v_{t_i,n}) \leq K, |q - t_i| \leq \lfloor L/2 \rfloor\} \quad (4)$$

Here, K denotes the dimension of the spatial kernel, L signifies the extent of the temporal kernel, and $d(\cdot, \cdot)$ characterizes the spacing between two nodes. The application of the STGC operation on a node $v_{t_i,n}$ can be articulated through the following equation:

$$f_{out}(v_{t_i,n}) = \frac{1}{Z_{t_i,n}} \sum_{v_{q,m} \in B(v_{t_i,n})} f_{in}(v_{q,m}) \cdot w(v_{q,m}) \quad (5)$$

Among them $Z_{t_i,n}$ is the normalization factor, $w(\cdot)$ is the spatiotemporal convolution kernel[19]. Here, the spatiotemporal convolution kernel $w(\cdot)$ is approximated by decomposing it into spatial convolution kernel $W_S \in R^{C \times M}$ and temporal convolution kernel $W_T \in R^{M \times L}$. Specifically, assuming that $f_{t_i} \in R^{N \times C}$ represents the input features of n ROIs at time point t_i in channel C and $A \in R^{N \times N}$ is a correlation matrix, the spatial convolution at time point t_i is defined as:

$$f'_{t_i} = \Lambda^{\frac{1}{2}}(A + I) \Lambda^{\frac{1}{2}} f_{t_i} W_s \quad (6)$$

Where $\Lambda^{n,n} = \sum_j A^{n,j} + I$ is a diagonal matrix; $f_{t_i} \in R^{N \times M}$ indicates the output characteristics of the M channel. This article further performs temporal convolution on the obtained spatial features. Taking nodes v_n as an example, the output after temporal convolution can be represented as:

$$f_n \otimes W_T \in R^{M \times S} \quad (7)$$

Among them $f_n \in R^{M \times S}$ are the features defined by nodes v_n on the timeline graph, $\otimes$ representing a standard one-dimensional convolution.

In this study, three tiers of STGC units are incorporated into each ST-GCN. The initial layer receives the C -channel feature as input, and each subsequent layer produces the M -channel feature as output. The output from the final STGC tier is then directed to a global average pooling layer, which yields a 64-dimensional graph-level representation vector. Concisely, this representation vector can be algebraically represented as:

$$z_i = F(X_i) \in R^{64} \quad (8)$$

Among them $F(\cdot)$ denotes the transformation function of the ST-GCN feature extraction component.

MLP predictor. For each view (e.g. X_1), this article also defines an MLP predictor $H(\cdot)$ for feature abstraction, which is applied to convert the output from one perspective or branch and align it with the other branch. Each MLP module comprises two dense layers, which contain 32 and 64 neurons, respectively. Taking $z_i (i = 1, 2)$ as input, the MLP predictor will generate a 64-dimensional feature vector:

$$p_i = H(z_i) = H(F(X_i)) \quad (9)$$

C. Contrastive Learning Module

Compare the loss function. Due to the representation vectors z_1 and p_2 of two different views from the same object, it is posited that they exhibit similarity within the new embedding space. Consequently, the objective is to reduce the negative correlation between the two resulting vectors z_1 and p_2 as outlined below:

$$\psi(z_1, p_2) = -\frac{z_1}{\|z_1\|_2} \cdot \frac{p_2}{\|p_2\|_2} \quad (10)$$

Where $\|\cdot\|_2$ represents a binomial[20]. For these two perspectives, we establish the following symmetrical loss function to enhance the resemblance between the two entities:

$$\xi = \frac{1}{2}\psi(z_1, p_2) + \frac{1}{2}\psi(z_2, p_1) \quad (11)$$

Gradient stopping technique. In line with the approach described by Chen and He[21], this study also employs a gradient stopping operation $g(\cdot)$ on each view within the pre-trained model, which is a crucial step to avoid model degradation. Consequently, equation (10) can be restated as follows:

$$\psi(g(z_1), p_2) \quad (12)$$

z_1 is considered a constant. Therefore, equation (11) can be rephrased as follows:

$$\xi = \frac{1}{2}\psi(g(z_1), p_2) + \frac{1}{2}\psi(g(z_2), p_1) \quad (13)$$

From equation (13), it can be observed that the ST-GCN feature encoder is applied to X_1 did not receive a gradient from z_1 in the first term, but received a gradient from p_1 in the first term; X_2 is also the case. This helps partially solve the problem of model crashes.

D. Mixup based Universum samples Module

Inspired by Universum learning, this paper proposes a novel method that combines contrastive learning with Mixup. The core concept is that mixed samples can serve as challenging negative examples. This article randomly blend two images from different categories, creating visually ambiguous data that is difficult to classify. We view the blending of images as a form of generalization and incorporate this into the contrastive loss function as negative samples across all datasets. Unlike traditional negative samples limited to specific anchor groups, this method uses Mixup to generate Universum samples that are distinct from the target class anchors. For each anchor, a comparison sample from another class is used to create Universum data points, which help define clearer boundaries between instances and classes. To address the issue of semantically similar samples from traditional Mixup strategies, the paper sets the Mixup parameter to a constant, generating Universum data that represents the target class within the data space.

This article uses enhanced sample data from two subjects, Y_1 and Y_2 , as the original data. Drawing inspiration from Universum learning, the Universum-style Mixup approach is employed to enhance the effectiveness of contrastive learning by introducing additional Universum examples. This method helps reduce visual ambiguity and facilitates clearer distinction between different classes by utilizing the boundaries that separate them. Unlike traditional mixing methods, Universum mixing pairs each anchor sample $x_{i,j} (i = 1, 2, \dots, T; j = 1, \dots, N)$ in a training batch with its out-of-class negative samples $x_{\bar{i},j} (i = 1, 2, \dots, T; j = 1, \dots, N)$ to create a Universum sample $u_{i,j}$. In contrast to traditional mixing strategies, which randomly sample mixing parameters from a β distribution, the method in this article sets the mixing parameter λ to a specific number. This approach reduces the chance of Universum data overlapping with the target class in the data space, ensuring that the mixed Universum data remains negative. The Universum data is generated through the following procedure:

$$u_{i,j} = \lambda \cdot x_{i,j} + (1 - \lambda)x_{\bar{i},j} \quad (14)$$

If $x_{i,j} \in Y_1$, then $x_{\bar{i},j}$ is any data point taken from Y_2 , λ is the Mixup parameter. The rest of this article, $u_{i,j}$ will be referred to as the "Universum sample" and $x_{i,j}$ used as its anchor point.

E. Universum Loss Module

Based on the window-slicing data augmentation method, this article uses a strategy to mix the enhanced data Y_1 and Y_2 from different subjects, generate data X_3 , and use it as the Universe sample data. In this article, the enhanced data $X_i (i = 1, 2)$ will be used for cosine similarity calculation with X_3 . The loss function is represented as:

$$L_{Uni} = \frac{1}{4}(\psi(g(z_3), p_1) + \psi(g(z_1), p_3) + \psi(g(z_3), p_2) + \psi(g(z_2), p_3)) \quad (15)$$

In the above situation, this article combines the loss functions ξ of the enhanced data $X_i (i = 1, 2)$ and the data X_3 generated by the method from different subjects to form the final loss function L_{sum} , namely:

$$L_{sum} = \frac{1}{6}(\psi(g(z_1), p_2) + \psi(g(z_2), p_1) - \psi(g(z_3), p_1) - \psi(g(z_1), p_3) - \psi(g(z_3), p_2) - \psi(g(z_2), p_3)) \quad (16)$$

In this function, minimizing the loss function aims to bring positive sample pairs closer together and separate Universum sample pairs.

TABLE I. Tested Cross-Site Categorization Model from NYU Site to UM Site on ABIDE

Method	ACC	AUC	REC	PRE	F1
CNN	64.1	70.2	55.3	59.1	54.2
GCN	66.1	72.1	57.2	63.1	59.2
ST-GCN	69.1	72.1	55.1	74.1	62.1
w/oMixup	63.1	65.1	63.2	68.1	65.1
Uni-GCL	83.3	93.6	86.0	87.5	84.9

The bold face denotes the best results

TABLE II. Tested Cross-Site Categorization Model from NYU Site to LEUVEN Site on ABIDE

Method	ACC	AUC	REC	PRE	F1
CNN	62.1	66.2	60.2	62.1	61.2
GCN	66.1	73.2	49.1	68.0	56.1
ST-GCN	71.2	66.2	62.2	72.2	66.2
w/oMixup	62.1	66.2	60.2	62.1	61.2
Uni-GCL	77.7	97.2	70.1	89.2	74.2

F. Fine-tune the model

The research aims to enhance the model's ability to analyze target fMRI data and apply a pre-trained source domain model to classification tasks. The refined model includes an ST-GCN feature extractor, two dense layers in MLPs, and a Softmax prediction layer. It uses partially labeled target fMRI data for supervised training, generating spatiotemporal maps and extracting features with ST-GCN. Brain disease classification is performed using the MLP and Softmax layers, with the model refined through cross-entropy loss minimization.

V. EXPERIMENTS

A. Experimental settings

In this study, five metrics were employed to assess the efficacy of various methods: accuracy (ACC), the area under the curve of the receiver operating characteristic (AUC), recall (REC), precision (PRE), and the F₁-score. The research employed a cross-site experimental design using the ABIDE dataset for autism spectrum disorder classification. The model was initially pre-trained in an unsupervised manner at the NYU site, followed by fine-tuning and evaluation at the UM and

LEUVEN sites respectively. During the fine-tuning phase, a 5-fold cross-validation strategy was utilized, where 80% of the samples from the target site/dataset were used for fine-tuning, and the remaining 20% were reserved for testing. The average and standard deviation of the outcomes were documented.

B. Cross-site classification results

For ABIDE, we pre-trained the Uni-GCL model at the NYU site, then fine-tuned and tested it at the UM and LEUVEN sites. In TABLE I and TABLE II, we reported the classification results of HC and ASD using Uni-GCL, Mixup free method, Convolutional Neural Network (CNN), Graph Convolutional Network (GCN), and Spatiotemporal Graph Convolutional Network (ST-GCN), respectively. From TABLE I and TABLE II, we can observe that the overall performance of Uni-GCL is superior to deep learning methods when using a Mixup strategy. The possible reason is that our Uni-GCL introduces universum sample data through the Mixup strategy, improving the robustness and generalization ability of the model, and optimizing it through unsupervised contrastive learning. It can extract richer and more transferable functional magnetic resonance imaging features without any label information, improving the classification performance of downstream tasks.

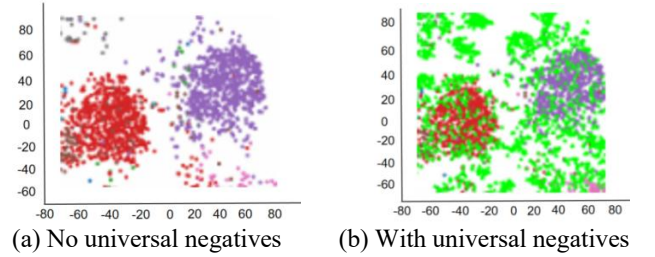


Fig 2. Uni-GCL Data Visualization

C. Visualization Analysis

This study extracts features from ABIDE test data to obtain two-dimensional visualization. Fig 2 (a) shows the feature distribution of training data points, while Fig 2 (b) adds the distribution of artificially synthesized universum negative features. For visualization purposes, this article specifically synthesized 10,000 single-sum data points from the test data with a mixed parameter of 0.5 and applied them to the Uni-GCL model. The model in this study can separate different clusters with larger boundaries in the embedding space. In Uni-GCL, a large amount of universum data is placed on the edges between different star clusters. Although the negative value of universum is a mixture of two different images of the same class, it cannot be assigned to either of these two classes; otherwise, it will cause negative distortion in the manifold space, where the decision boundary may learn towards one of the classes. By moving universum data to the edge space, it is possible to better separate different classes and draw clearer decision boundaries.

D. Sensitivity Analysis

This study tests different λ selected for mixed use: 0.3, 0.4, 0.5, 0.6 and 0.7, respectively. As shown in Fig 3 below, the model performs best when two images are evenly mixed to

produce a single negative value. This result is consistent with our intuition; when the contribution of two images to their mixed image is equal, the mixed image is semantically farthest from its original image.

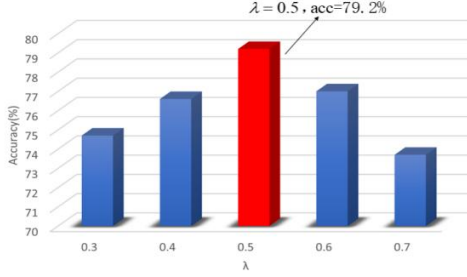


Fig 3. Sensitivity analysis of λ value

Here, a concise analysis is provided to illustrate the importance of setting the value of λ to 0.5. In the context of Mixup, when blending data points with a mixing coefficient of λ , the resulting label should be $y = \{0, \dots, \lambda, \dots, 1 - \lambda, \dots, 0\}$, where the placement of λ and $1 - \lambda$ corresponds to the respective mixed data. Consequently, the information entropy of y can be computed as follows:

$$H(y) = -\lambda \log(\lambda) - (1 - \lambda) \log(1 - \lambda) \quad (17)$$

To obtain the maximum value of $H(\lambda)$, we derived the gradient relative to λ of $H(\lambda)$:

$$\frac{\partial H(y)}{\partial \lambda} = \log\left(\frac{1-\lambda}{\lambda}\right) \quad (18)$$

Obviously, at that time $\lambda = 0.5$, $\partial H(y)/\partial \lambda = 0$. At this juncture, the maximum value of $H(y)$ is achieved, and the label vector y exhibits the greatest degree of uncertainty. Therefore, at that time $\lambda = 0.5$, the data based on Mixup could be best described as the universal class of all data, since such samples are the most uncertain.

VI. CONCLUSION

This study employs a window-slicing strategy for data augmentation and introduces universum samples to reduce dependence on a large amount of sample data. Positive sample data is extracted through unsupervised contrastive learning for feature extraction, while a mixed strategy generates data used as negative samples. By combining two types of loss functions, the pre-trained model is transferred to downstream tasks for initializing and updating fine-tuning models to improve model performance. The experimental results indicate that introducing universum samples can effectively improve the performance of brain networks and significantly reduce the reliance for large sample data in contrastive learning.

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一、EI 4条

1. Universum Inspired Graph Contrast Learning for Autism Spectrum Disorder Identification

Accession number: 20250517800835

Authors: Qin, Hao (1); Xu, Xiaowen (2, 3); Li, Weikai (1, 4)

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Publisher: Institute of Electrical and Electronics Engineers Inc.

Abstract: The functional brain network (FBN) has recently emerged as a pivotal biomarker in the diagnosis of autism spectrum disorder (ASD). Meanwhile, graph convolutional networks (GCNs) represent a promising strategy for the concurrent extraction of FBN features and the facilitation of ASD identification. Unfortunately, the efficacy of existing GCN methods is often limited by their reliance on extensive annotated datasets. To address this limitation, we propose the Universum Inspired Graph Contrast Learning (Uni-GCL) framework, a novel approach to brain disease analysis that capitalizes on the generation of Universum samples for the pre-training of the GCN model. Specifically, we use a Mixup technique to generate universum samples and design a universum-inspired contrastive loss function. This methodology necessitates only a modest number of samples for finetuning the pre-trained model, effectively reducing data requirements while simultaneously boosting model performance. To systematically assess the performance of Uni-GCL, we conducted rigorous experiments on the Autism Brain Imaging Data Exchange (ABIDE) initiative for ASD identification. The results demonstrate that Uni-GCL, augmented with universum samples, yields compelling outcomes, underscoring its potential as a robust method in the diagnostic arsenal for ASD. © 2024 IEEE.

Number of references: 21

Main heading: Diseases

Controlled terms: Convolutional neural networks - Diagnosis

Uncontrolled terms: Autism spectrum disorders - Brain networks - Convolutional networks - Functional brain network - Functionals - Graph contrast

learning - Graph convolutional network - Network features - Network
methods - Universum learning

Classification code: 102.1 Medicine - 102.1.2 Health Science - 1101.2.1 Deep Learning

DOI: 10.1109/ICCSI62669.2024.10799329

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2. Topology-Enhanced Multi-Modal Residual Graph Convolutional Network for Autism Spectrum Disorder Identification

Accession number: 20250517799832

Authors: Deng, Jipeng (1); Guan, Lihe (1); Li, Weikai (1, 2)

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Publisher: Institute of Electrical and Electronics Engineers Inc.

Abstract: Functional Brain Networks (FBNs) have shown promise in diagnosing neurological disorders, particularly Autism Spectrum Disorder (ASD). Currently, Graph Convolutional Networks (GCNs) excel in processing FBN data, making them instrumental in learning from functional Magnetic Resonance Imaging (fMRI) and analyzing ASD. Despite their advantages, existing methods primarily concentrate on the structural properties of brain networks, often overlooking the essential topological information. To bridge this gap, we introduce a Topologically Enhanced Multi-Pattern Residual Graph Learning (TEMPRGL) framework. Our approach begins by constructing diverse brain networks from fMRI data, capturing a range of connectivity patterns. We then employ a dual-feature extraction strategy on FBNs: one that utilizes multiple Residual GCN modules to generate multi-modal graph representations, which are subsequently integrated; and another focuses on combining topology information from FBN. To validate our TEMPRGL, we experiment on the Autism Brain Imaging Data Exchange (ABIDE) dataset. It attained an impressively high accuracy rate of 92.5%, underscoring its effectiveness in neurological disorder diagnosis. © 2024 IEEE.

Number of references: 29

Main heading: Convolutional neural networks

Controlled terms: Brain mapping - Diagnosis - Diseases - Graph neural networks - Magnetic resonance imaging - Network theory (graphs) - Neurons

Uncontrolled terms: Autism spectrum disorders - Brain networks - Convolutional

networks - Convolutional neural network - Functional connectivity
networks - Functional magnetic resonance imaging - Functionals - Multi
patterns - Multi-modal - Neurological disorders
Classification code: 101.1 Biomedical Engineering - 102.1 Medicine - 102.1.2 Health
Science - 103 Biology - 1101 Artificial Intelligence - 1101.2.1 Deep
Learning - 1201.8 Discrete Mathematics and Combinatorics, Includes Graph Theory, Set
Theory - 709 Electrical Engineering, Other Topics - 746 Imaging Techniques

Numerical data indexing: Percentage 9.25E+01%

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3. Multi-View Comparative Self-Supervised Graph Learning for Neuro-disease Diagnosis Using Functional Brain Networks

Accession number: 20250517800859

Authors: Tian, Nan (1); Xu, Xiaowen (2, 3); Li, Weikai (1, 4)

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Publisher: Institute of Electrical and Electronics Engineers Inc.

Abstract: The burgeoning interest in diagnosing neurological diseases through the analysis of functional brain networks (FBNs) has been a focal point of recent research. Despite the promise of this approach, conventional graph learning techniques are often constrained by their reliance on extensive datasets annotated with clinical information. To transcend this constraint, we introduce a novel self-supervised graph learning model, termed Multi-View Comparative Self-Supervised Graph Learning (MCS-GL), which leverages Personalized PageRank (PPR) to increase samples and uses local-global information as multi view for comparative learning to enhance FBN-based diagnostics. With this approach, a pre-trained model is designed to generate information-rich Functional Magnetic Resonance Imaging (fMRI) representations from unlabeled training data. Fine-tuning the model for downstream disease identification tasks significantly reduces the requirement for large amounts of labeled training data. We have rigorously evaluated the efficacy of our MCS-GL model through its application to an autism diagnosis task. The experimental outcomes not only substantiate the

superiority of our model but also underscore the pivotal role of self-supervised learning in advancing FBN-based diagnostics of neurological diseases. © 2024 IEEE.

Number of references: 19

Main heading: Self-supervised learning

Controlled terms: Contrastive Learning - Convolutional neural networks - Graph neural networks - Labeled data - Magnetic resonance imaging - Neurons - Semi-supervised learning

Uncontrolled terms: Autism disorders - Brain networks - Convolutional neural network - Disease diagnosis - Functional brain network - Functionals - Learning models - Multi-views - Neuro-disease diagnose - Neurological disease

Classification code: 103 Biology - 1101 Artificial Intelligence - 1101.2 Machine Learning - 1101.2.1 Deep Learning - 709 Electrical Engineering, Other Topics - 746 Imaging Techniques - 903.1 Information Sources and Analysis

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4. Enhancing Functional Brain Network Construction for Neurological Disorder Diagnosis through Modularity-Driven Regularization

Accession number: 20250517800848

Authors: Sun, Danjie (1); Chen, Xiaofeng (1); Li, Weikai (1, 2)

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Abstract: The burgeoning field of Functional Brain Network (FBN) analysis has emerged as a promising avenue for diagnosing neurological disorders. As innovative biomarkers, FBNs offer a novel perspective on probing the intricacies of brain connectivity. However, the fidelity of FBN construction depends on the quality of data acquisition, which is often a formidable challenge due to inherent limitations. Inspired by the no free lunch theorem, a straightforward ambition is to integrate the exogenous knowledge to enhance the fidelity of FBNs. Motivated by this ambition, this paper provides a Modularity-Driven Regularization (MDR) for the FBN construction. Specifically, we design and incorporate a dyadic L21 regulariser into the FBN model to formulate the modularity prior. Furthermore,

we have developed a generalized alternating multiplier method algorithm for the estimation of the FBN. To demonstrate the effectiveness of our MDR, we perform rigorous experiments on a Mild Cognitive Impairment (MCI) identification task using the NITRC dataset. The validity of our model has been confirmed by the experimental outcomes. Additionally, it enhances biomarker precision for diagnosing neurological disorders like MCI, supporting potential early intervention strategies. © 2024 IEEE.

Number of references: 31

Main heading: Neurons

Controlled terms: Diagnosis - Neural networks

Uncontrolled terms: Brain networks - Cognitive impairment - Functional brain network - Functionals - Generalized alternating multiplier method algorithm - Mild cognitive impairment - Modularity - Multipliers method - Network construction - Neurological disorders

Classification code: 101.1 Biomedical Engineering - 102.1 Medicine - 103 Biology - 1101 Artificial Intelligence

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